Jing Lan

jlan25@wisc.edu · homepage · LinkedIn · Madison, WI

Education

University of Wisconsin-Madison 2024 – (expected) 08/2028
Ph.D. in Computer Sciences
The Chinese University of Hong Kong 2022 – 2024
M.Phil. in Computer Science and Engineering
Sun Yat-sen University 2018 – 2022
B.Eng. in Computer Science, ***GPA:* 3.9 / 4.0**

Technical Skills

Infrastructure: Kubernetes, Docker, Ansible, Helm, Kueue, Prometheus, Grafana
Cloud / Development: Kubespray, DevSpace, Supabase, GitLab, Git, CI/CD, React, FastAPI
Systems: Linux, distributed systems, Supercomputers, networking, container orchestration
Languages: Python, C++, C, JavaScript, Shell

Research & Engineering Experiences

EFabric: Cloud-Native AI4Science Computing Platform — NetLab | 2025 – Present

Public platform: https://efabric.cs.wisc.edu
In-house Kubernetes platform for scientific computing, providing multi-tenant compute, experiment artifact management, secure cluster access, and petabyte-scale shared storage for domain researchers.

- Architected and deployed the platform across **75 bare-metal GPU servers**, owning the full lifecycle from hardware bring-up and cluster provisioning to service delivery and production operations.
- Built **reproducible GPU cluster provisioning** with Ansible and Kubespray, automating Kubernetes setup and scaling, firewall policy enforcement, NVIDIA GPU and monitoring software stacks, and Kueue scheduling configurations to accelerate node onboarding and minimize configuration drift.
- Established a **DevSpace-based full-stack development/deployment setup**; built a React frontend with BFF routing, a replicated FastAPI backend for experiment orchestration and management, MCP-based agentic tools, and an SSH gateway for secure remote workspace access.
- Designed and operated a **1.2 PB shared storage platform** with TiKV, SeaweedFS, and JuiceFS, exposing S3 and POSIX interfaces to cluster services and research workloads.
- Implemented **authentication, permission controls, and resource admission** with Supabase; integrated Kueue for queue-based resource partitioning and multi-tenant scheduling. Built **cluster-wide observability** with Prometheus and Grafana covering clusters, storage, servers, and experiments.

Flavius: Disaggregated Graph OLAP Engine — Husky Data Lab | 2022 – 2024

- Co-designed a **hybrid pull/push execution pipeline** for Flavius's disaggregated master/compute architecture, using staged dataflow to support complex graph queries with branching and iterative structures, while accelerating per-stage data processing with fine-grained parallel tasks.
- Designed and implemented an **adaptive two-level task scheduler** for concurrent graph queries, using batch size to control per-query resource consumption and task size to tune execution parallelism, balancing **throughput** and **P99 latency** under highly skewed mixed workloads.
- Built a **workload-aware scheduler tuner** using derivative-free coordinate descent and integrated it into a **C++ distributed query engine with 41K LOC**; Evaluated against Memgraph, TigerGraph and NebulaGraph, Flavius achieved the best P99 latency across LDBC and Amazon workloads, and up to **3.6× lower tail latency** than using fixed scheduling configurations.

Student Cluster Competitions (SCC) — Sun Yat-sen University | 2020 – 2022

- Optimized highly parallel HPC workloads across **heterogeneous platforms** including Tianhe-2, UofT Niagara, and Chameleon Cloud, with hands-on experience in cluster deployment, workload benchmarking, and performance tuning.
- Led the university HPC competition team from 2021 to 2022, organizing weekly technical sessions and practices on server installation, cluster management, scientific workload deployment, benchmarking, and **CUDA/OpenMP/MPI** optimization.
- Champion, IndySCC@ACM Supercomputing 2021; Honorable Mention, SCC@ISC High Performance 2021.